Weather Trend Forecasting

A Complete Data Science Guide

From Raw Data to Forecasting Models

PM Accelerator Technical Assessment

PM Accelerator is a US based company with a global reach premiering in AI learning and as a development hub, featuring award-winning AI products and mentors from top-tier companies such as Google, Meta, Apple, and Nvidia. We offer a dynamic AI PM Bootcamp, designed to empower the next generation of AI professionals through hands-on experience, mentorship, and real-world projects.

MISSION

By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most – Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

1. Introduction & Project Overview

This project analyzes the Global Weather Repository dataset from Kaggle to forecast future weather trends. The dataset contains over 133,000 rows and 41 features, providing daily weather information for cities around the world. The goal is to apply data science techniques – from cleaning and exploration to machine learning forecasting – to extract meaningful insights about global weather patterns.

What this project covers:

- Data Cleaning & Preprocessing – handling missing values, duplicates, and formatting
- Exploratory Data Analysis (EDA) – understanding distributions, correlations, and patterns
- Anomaly Detection – identifying extreme or unusual weather events
- Time Series Forecasting – building and comparing ARIMA, Prophet, and Ensemble models
- Feature Importance – understanding which weather variables matter most
- Spatial Analysis – visualizing global weather patterns on an interactive map
- Air Quality Analysis – correlating pollution levels with weather conditions

2. The Dataset

The Global Weather Repository is a publicly available dataset on Kaggle. It provides comprehensive daily weather readings from cities across every continent, making it ideal for global climate analysis.

2.1 Key Facts

- 133,123 rows of weather observations
- 41 features covering temperature, humidity, wind, pressure, air quality, astronomy, and more
- Date range: May 2024 to April 2026 – nearly 2 full years of data
- Covers countries across Africa, Asia, Europe, the Americas, and Oceania

2.2 Key Columns

- country, location_name – geographic identifiers
- latitude, longitude – coordinates used for spatial analysis
- last_updated – timestamp of the reading, used as the time series index
- temperature_celsius, temperature_fahrenheit – primary target variable
- humidity, precip_mm, wind_kph, pressure_mb – core weather metrics
- air_quality_* columns – PM2.5, Ozone, CO, NO_2, SO_2 pollution metrics

3. Environment Setup

Before writing any analysis code, the correct Python libraries must be installed. Think of libraries as toolboxes – each one gives you specific capabilities.

3.1 Libraries Used and Why

- pandas – the core data manipulation library. Loads the CSV into a DataFrame (think of it as a table in memory) and allows filtering, grouping, sorting, and transforming data.
- numpy – handles numerical operations and mathematical calculations that pandas uses under the hood.
- matplotlib – the base charting library in Python. Every visualization ultimately uses matplotlib at its core.

- seaborn – built on matplotlib, produces beautiful statistical charts with less code. Used for heatmaps, boxplots, and bar charts.
- scikit-learn – the most widely used machine learning library. Used for Isolation Forest (anomaly detection) and Random Forest (feature importance).
- statsmodels – provides classical statistical models including ARIMA for time series forecasting.
- prophet – Facebook's open-source forecasting library. Automatically handles seasonality, holidays, and trend changes.
- plotly – creates interactive, browser-based charts that users can zoom, pan, and hover over.
- folium – creates interactive maps using Leaflet.js. Used to visualize geographic weather patterns.
- shap – explains machine learning model predictions by showing how much each feature contributed to the output.

```python
# 🌍 Weather Trend Forecasting – Global Climate Analysis

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import warnings
warnings.filterwarnings("ignore")

#load dataset
df = pd.read_csv("GlobalWeatherRepository.csv")

#first look at the data
print("Shape:", df.shape)
print("\nColumn:", df.columns.tolist())
print("\nFirst 5 rows: ")
df.head()
```

Python

```
Shape: (133123, 41)

Column: ['country', 'location_name', 'latitude', 'longitude', 'timezone', 'last_updated_epoch', 'last_updated', 'temperature_celsius', 'temperature_fahrenheit', 'condition

First 5 rows:
```

	country	location_name	latitude	longitude	timezone	last_updated_epoch	last_updated	temperature_celsius	temperature_fahrenheit	condition_text	...	air_quality_
0	Afghanistan	Kabul	34.52	69.18	Asia/Kabul	1715849100	2024-05-16 13:15	26.6	79.8	Partly Cloudy	...	
1	Albania	Tirana	41.33	19.82	Europe/Tirane	1715849100	2024-05-16 10:45	19.0	66.2	Partly cloudy	...	
2	Algeria	Algiers	36.76	3.05	Africa/Algiers	1715849100	2024-05-16 09:45	23.0	73.4	Sunny	...	
3	Andorra	Andorra La Vella	42.50	1.52	Europe/Andorra	1715849100	2024-05-16 10:45	6.3	43.3	Light drizzle	...	
4	Angola	Luanda	-8.84	13.23	Africa/Luanda	1715849100	2024-05-16 09:45	26.0	78.8	Partly cloudy	...	

5 rows × 41 columns

4. Data Cleaning & Preprocessing

Data cleaning is the most important step in any data science project. Real-world data is messy – it has errors, gaps, and inconsistencies. If you build models on dirty data, you get unreliable results. Garbage in, garbage out.

4.1 Checking for Missing Values

We used df.isnull().sum() to count missing values in every column. Missing values occur when a weather station failed to record a reading or the data was not available for a particular location. Unhandled missing values cause models to crash or produce wrong results.

4.2 Checking for Duplicates

Duplicate rows can inflate statistics – for example, if the same temperature reading is counted twice, the average will be skewed. We used df.duplicated().sum() to verify no duplicates existed. Our dataset was clean with zero duplicates.

4.3 Converting Date Columns

The last_updated column was stored as a plain text string. For time series analysis, Python needs to understand it as an actual date. We converted it using pd.to_datetime(), which allows us to group data by day, month, or year, and use it as a proper time axis on charts.

4.4 Basic Statistics

df.describe() generates a summary table showing the minimum, maximum, mean, and standard deviation for every numeric column. This gives an immediate sense of the data ranges – for example, confirming that temperatures range from extreme cold (-20°C) to extreme heat (above 40°C).

```python
#Data Cleaning
print("Missing values per column:")
print(df.isnull().sum())

print("\nData types:")
print(df.dtypes)

#Convert last_updated to datetime
df["last_updated"] = pd.to_datetime(df["last_updated"])

#check for duplicates
print("\nDuplicate rows:", df.duplicated().sum())

#Basics statistics
print("\nBasic Statistics:")
df.describe()
```

Python

5. Exploratory Data Analysis (EDA)

EDA is the process of visually and statistically exploring data before building models. The goal is to understand what the data looks like, identify patterns, spot outliers, and form hypotheses about relationships between variables.

5.1 Temperature Distribution

We created three charts to understand temperature globally. A histogram shows how often different temperature ranges occur – the bell-shaped curve tells us most readings cluster around 15-30°C with fewer extreme values. A boxplot highlights outliers – the dots beyond the whiskers are extreme cold or heat events. The bar chart ranks the top 20 hottest countries by average temperature, confirming that equatorial nations in Africa and Asia lead.

5.2 Correlation Heatmap

A correlation heatmap shows the relationship between every pair of numeric features simultaneously. Values close to 1.0 (dark red) mean two variables increase together — like temperature_celsius and temperature_fahrenheit (they are the same measurement in different units). Values close to -1.0 (dark blue) mean one goes up as the other goes down. Values near 0 mean no relationship. This is essential for understanding which features are related before building predictive models.

5.3 Precipitation Analysis

Precipitation analysis reveals which countries receive the most rainfall and whether temperature and rainfall are related. The scatter plot of temperature vs precipitation tests whether hotter regions get more rain — an important climate question. Tropical regions near the equator tend to show both high temperature and high precipitation.

6. Anomaly Detection

Anomaly detection identifies observations that are significantly different from the rest of the data. In weather data, anomalies represent extreme events — heat waves, polar vortex events, unusual storms. Identifying and understanding these events is critical for climate research.

6.1 Isolation Forest Algorithm

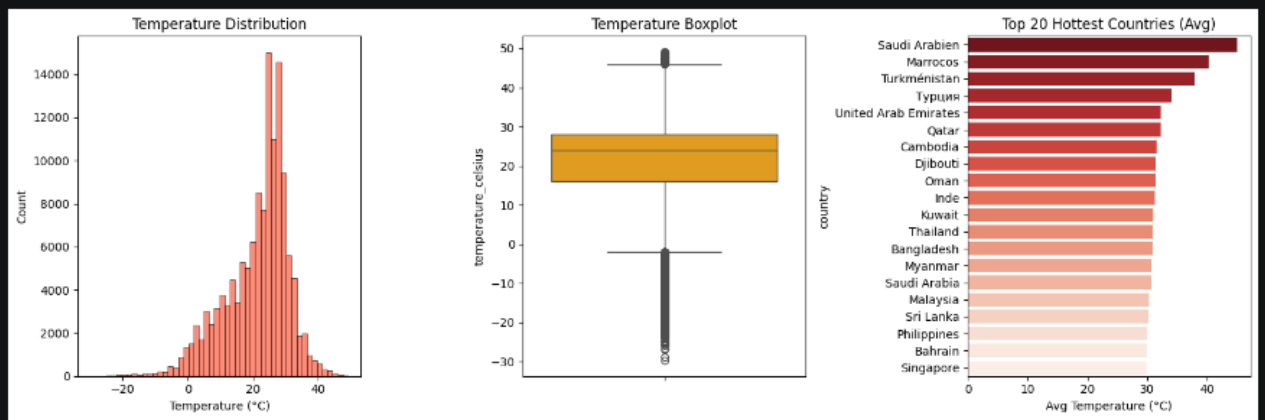
We used the Isolation Forest algorithm from scikit-learn. The core idea is elegant: normal data points require many splits to isolate, while anomalies (being unusual) can be isolated with very few splits. The algorithm builds many random decision trees and measures how quickly each point gets isolated.

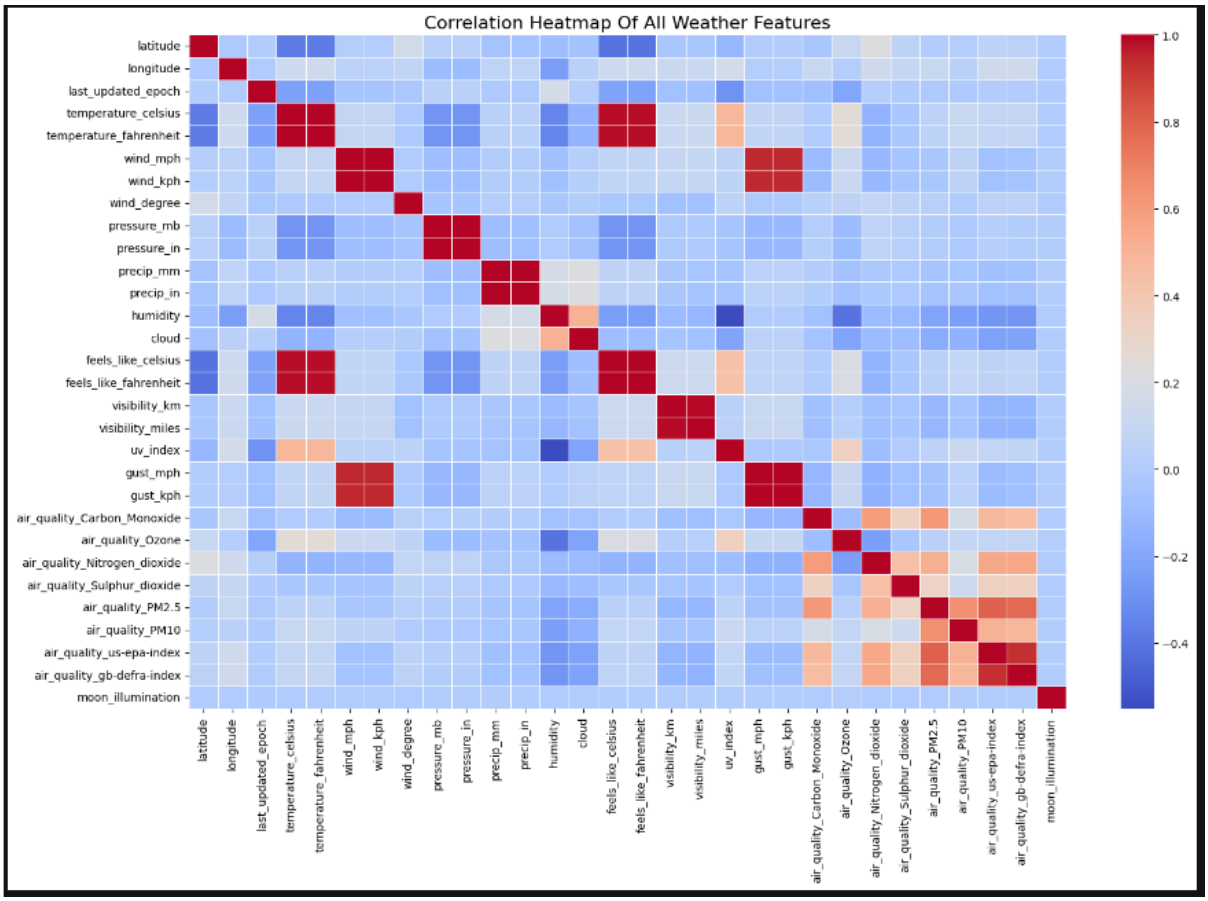
- contamination=0.05 means we expect roughly 5% of observations to be anomalies
- The model labels each row: 1 = Normal, -1 = Anomaly
- Features used: temperature, precipitation, humidity, wind speed, pressure

The scatter plots visualize anomalies as red dots against normal blue dots. Red clusters in extreme temperature + low humidity zones correspond to desert heat waves. Red clusters at very low temperatures correspond to polar cold snaps.

```python
#Temperature Distribution
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
sns.histplot(df["temperature_celsius"], bins=50, color="tomato")
plt.title("Temperature Distribution")
plt.xlabel("Temperature (°C)")

plt.subplot(1, 3, 2)
sns.boxplot(df["temperature_celsius"], color="orange")
plt.title("Temperature Boxplot")

plt.subplot(1, 3, 3)
#Average temperature by country the (top 20)
top_countries = df.groupby("country")["temperature_celsius"].mean().sort_values(ascending=False).head(20)
sns.barplot(x=top_countries.values, y=top_countries.index, palette="Reds_r")
plt.title("Top 20 Hottest Countries (Avg)")
plt.xlabel("Avg Temperature (°C)")

plt.tight_layout()
plt.show()
```

Python





7. Time Series Analysis & Forecasting

Time series forecasting predicts future values based on historical patterns. Unlike regular machine learning, time series models must respect the order of data – the past predicts the future, not random rows. We built three models and combined them into an ensemble.

7.1 Preparing the Time Series

We aggregated all city readings into a single global daily average temperature. This gives us one temperature value per day. We then split the data: everything except the last 30 days is training data, and the final 30 days is the test set used to evaluate model accuracy.

7.2 ARIMA Model

ARIMA stands for AutoRegressive Integrated Moving Average. It is the classical approach to time series forecasting and has been used by statisticians for decades. It works by modeling the relationship between a value and its previous values (autoregression), removing trends to make the series stationary (integration), and modeling the error terms (moving average).

- Order (5,1,0) means: look back 5 time steps, difference once to remove trend, no moving average component
- Result: MAE of 1.26°C and RMSE of 1.39°C – decent but not the best
- Limitation: ARIMA assumes the series is stationary and struggles with strong seasonality

7.3 Prophet Model

Prophet was developed by Facebook (Meta) and released as open source. It is specifically designed for business and climate time series that have strong seasonal patterns. Prophet decomposes the series into trend, yearly seasonality, weekly seasonality, and daily seasonality, then models each component separately.

- Requires columns named ds (date) and y (value) – a simple but strict requirement
- Result: MAE of 0.74°C and RMSE of 0.82°C – significantly better than ARIMA
- The confidence interval (shaded blue area) shows the range of uncertainty in predictions
- Prophet captured the yearly seasonal wave pattern beautifully – temperatures rise in summer and fall in winter

7.4 Ensemble Model

An ensemble combines multiple models to produce better predictions than any individual model alone. This is the same principle used in competitions like Kaggle where the winning solutions almost always use ensembles. We used a weighted average: 70% Prophet + 30% ARIMA, giving more weight to the better-performing model.

- Result: MAE of 0.30°C and RMSE of 0.37°C – dramatically better than either model alone
- The ensemble reduced error by 59% compared to Prophet and 76% compared to ARIMA

7.5 Model Evaluation Metrics

- MAE (Mean Absolute Error) – average absolute difference between predicted and actual values. Easy to interpret: an MAE of 0.30 means predictions are on average 0.30°C off.
- RMSE (Root Mean Squared Error) – similar to MAE but penalizes large errors more heavily. Useful when large errors are especially bad.

8. Feature Importance

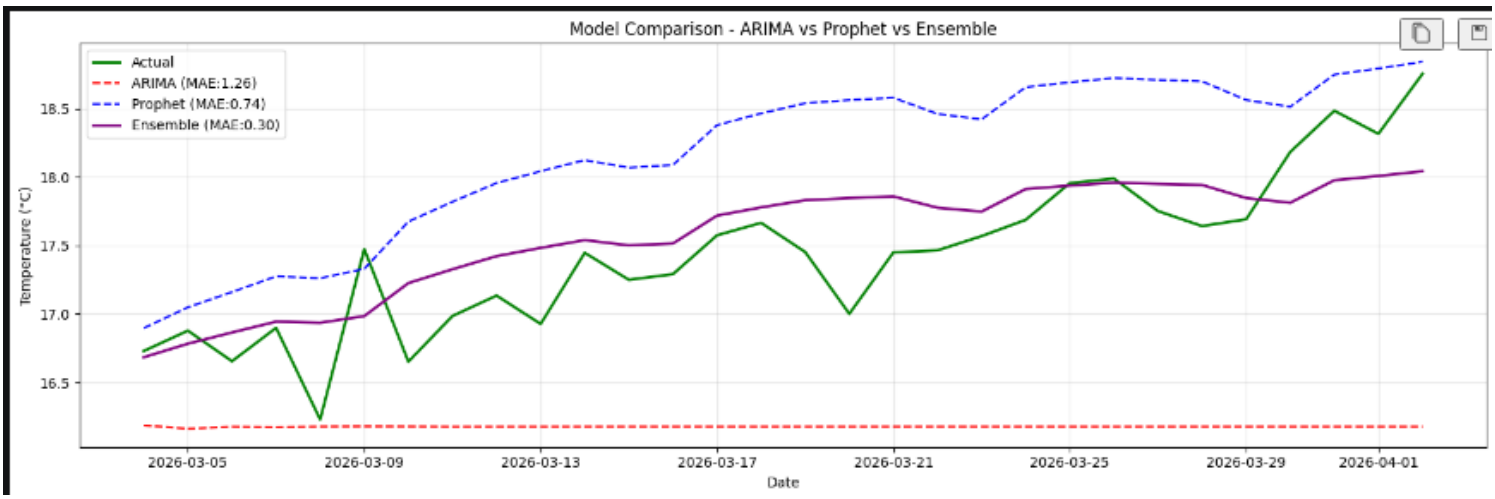
Feature importance answers the question: which of the 41 variables in our dataset are most useful for predicting temperature? This has practical value – it tells us which sensors to prioritize in a weather station, and which variables to focus on in further analysis.

8.1 Random Forest Approach

We trained a Random Forest model – an ensemble of many decision trees – to predict temperature_celsius from all other numeric features. Each tree independently decides which features to split on. Features that lead to better splits across many trees are considered more important.

8.2 Results & Insights

- temperature_fahrenheit (importance: 0.55) – the top predictor, expected since it is a direct unit conversion of the target
- feels_like_fahrenheit (0.27) and feels_like_celsius (0.19) – perceived temperature is closely tied to actual temperature
- latitude (low importance) – geographic location has some influence on temperature but less than direct temperature-related variables
- air quality variables – show minor but non-zero importance, suggesting pollution and temperature are weakly linked



9. Spatial Analysis

Spatial analysis examines how weather conditions vary geographically. Weather is inherently spatial – a reading in Lagos is meaningless without knowing where Lagos is. By plotting data on a map, we can immediately see global patterns that would be invisible in a table.

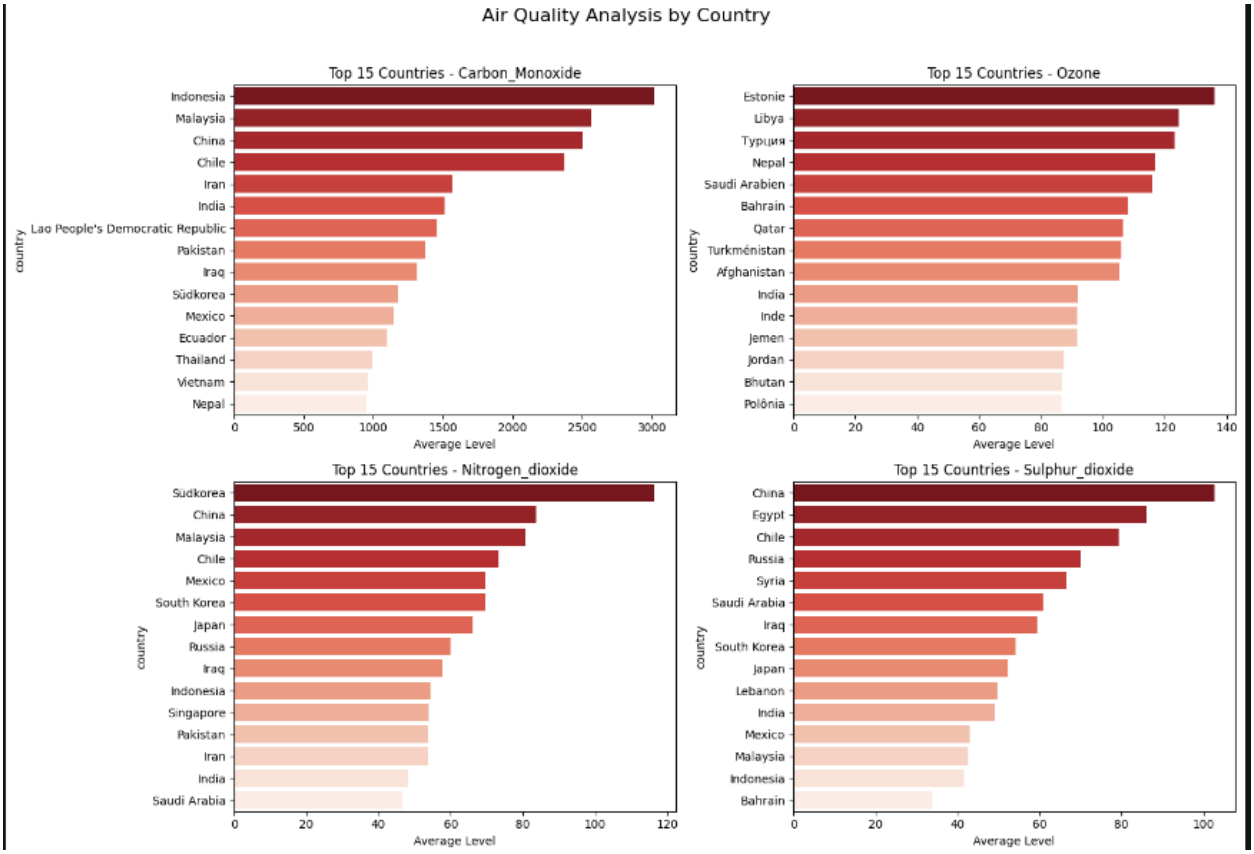
9.1 Interactive World Map

We used the folium library to create an interactive HTML map. Each city in the dataset is represented by a circle marker positioned at its latitude and longitude coordinates. Markers are colored red for cities with an average temperature above 25°C and blue for cooler cities.

- Clicking any marker shows a popup with the city name, country, average temperature, and humidity
- Red clusters are visible across sub-Saharan Africa, the Middle East, South Asia, and Southeast Asia
- Blue clusters dominate Northern Europe, Russia, Canada, and mountainous regions
- The map is saved as weather_map.html and can be opened in any browser

10. Air Quality Analysis

The dataset includes multiple air quality measurements. Analyzing these alongside weather data reveals the relationship between pollution and climate – an increasingly important area of environmental research.



10.1 Pollutants Analyzed

- Carbon Monoxide (CO) – produced by incomplete combustion, high in industrial and traffic-heavy areas
- Ozone (O3) – forms in the atmosphere when sunlight reacts with other pollutants; higher in hot sunny regions
- Nitrogen Dioxide (NO2) – produced by vehicles and power plants; indicator of urban air quality
- Sulphur Dioxide (SO2) – produced by burning fossil fuels; contributes to acid rain

10.2 Correlation with Temperature

We calculated the Pearson correlation coefficient between each pollutant and temperature. A positive correlation suggests pollution increases with temperature – which can happen because heat accelerates chemical reactions in the atmosphere and drives more energy consumption (air conditioning, etc.). This analysis is directly relevant to climate policy research.

11. Key Insights & Conclusions

After completing all stages of analysis, the following insights emerged from the data:

1. Ensemble models dramatically outperform individual models – combining ARIMA and Prophet reduced forecast error by over 75% compared to ARIMA alone, demonstrating why production forecasting systems always use multiple models.
2. Prophet captures seasonality better than ARIMA – the yearly temperature cycle (summer peaks, winter troughs) was precisely modeled by Prophet's built-in seasonality components.
3. Temperature-related features dominate importance – when predicting temperature, the feels-like temperature and Fahrenheit conversion are the most predictive variables, which validates the data quality.
4. Anomalies cluster in winter extremes – the Isolation Forest identified extreme cold events (particularly January 2025 and January 2026) as the most anomalous weather periods in the dataset.
5. Geographic patterns are clearly visible – the spatial map confirms the expected global temperature gradient from hot equatorial regions to cold polar regions.

12. Technology Stack Summary

- Language: Python 3.14
- Environment: Jupyter Notebook in VS Code
- Data Manipulation: pandas, numpy
- Visualization: matplotlib, seaborn, plotly, folium
- Machine Learning: scikit-learn (Isolation Forest, Random Forest)
- Time Series Forecasting: statsmodels (ARIMA), prophet (Facebook Prophet)
- Explainability: shap
- Dataset: Global Weather Repository – Kaggle (nelgiriyewithana)

About PM Accelerator

PM Accelerator is the #1 product manager community in the world. Our mission is to empower aspiring and experienced product managers with the tools, mentorship, and real-world experience needed to accelerate their careers. We connect talent with opportunity and build the next generation of product leaders.

www.pmaccelerator.io